

ERIC HAN

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EDUCATION

Stony Brook University

2025-Present

Ph.D. in Computational Applied Mathematics

Overall GPA: 3.81

Boston University

2023-2025

B.A. in Mathematics, Applied

Overall GPA: 3.87

CUNY Baruch College

2021-2023

B.B.A. in Statistics and Quantitative Modeling

Minor in Advanced Business Analytics and Mathematics

Overall GPA: 3.97

RESEARCH AND ACADEMIC EXPERIENCES

Air Force Research Lab

Summer 2026

Visiting Faculty Researcher

I am developing novel reinforcement learning techniques to discover optimal network topologies for distributed network applications.

Neuronal Dynamics at Boston University

June 2024 - Present

Undergraduate Researcher

Advisors: Caitlin Lienkaemper, Boston University; Gabriel Ocker, Boston University

I investigated the stability of high-dimensional coupled dynamical systems, aiming to identify a critical phase transition boundary between global fixed-point stability and chaotic attractors in large networks. The existing deterministic framework could not analyze random topologies. I developed a Python simulation to solve non-linear ODEs on parameterized random graphs, later ported to Julia for faster runtime. To analyze large state spaces, I optimized the algorithm by vectorizing the core numerical integration, using native linear algebra algorithms, and implementing multi-core parallelization on my GPU. This enabled efficient parameter sweeps across networks to analyze emergent dynamical regimes. This optimization increased simulation capacity tenfold (from $N=1,000$ to $N=10,000$). The resulting heatmaps empirically located the stability boundary, supporting my analytical derivation of the phase transition formula. I am currently preparing a manuscript for submission to math-physics journals.

Polymath Jr. at CUNY Baruch College

June 2023 - August 2023

Undergraduate Research Assistant

Advisors: Matthew Junge, CUNY Baruch College; Si Tang, Lehigh University

I addressed an open conjecture regarding the dynamical behavior of stochastic particle systems on infinite branching networks. I aimed to determine the precise upper bounds of the "drift" parameter governing the phase transition between recurrence and transience. I developed a rigorous numerical verification framework, engineering a recursive search algorithm that utilized Sturm's theorem to verify polynomial root locations with exact precision. I significantly optimized the solver's computational efficiency by implementing an inductive robustness distribution construction scheme. This optimization enabled me to scale the feasible analysis range from simple structures with out-degree 2 to high-degree systems ($d = 50$), while allowing for the rigorous development of bounds up to $d = 13$. These high-precision numerical results provided the necessary evidence to support a theoretical proof of asymptotic monotonicity, which was subsequently published in a peer-reviewed article in *Electronic Communications in Probability*.

TECHNICAL PROJECTS

Numerical Linear Algebra Algorithm Development	<i>January 2026 - May 2026</i>
Implemented and benchmarked core numerical linear algebra algorithms (eigenvalue solvers, factorizations) in C and Python to rigorously evaluate algorithmic complexity and machine precision limits. Benchmarked custom implementations against standard LAPACK libraries with both well- and ill-conditioned matrices to validate computational accuracy, numerical stability, and exactness of the algorithms.	
Finite Difference Solver Development	<i>January 2026 - May 2026</i>
Engineered multi-dimensional finite difference simulators in Python and C to model highly non-linear physical systems, mainly focusing on conservative, entropy-preserving methods for shockwave problems. Extended numerical schemes using higher-order discretizations and TVD-property methods to suppress oscillations near discontinuities, validating accuracy and stability via von Neumann analysis and norm-based convergence.	
Physics Informed Neural Networks for Inverse PDEs	<i>November 2025 - December 2025</i>
Implemented Physics-Informed Neural Networks (PINNs) in PyTorch to solve nonlinear PDEs. Analyzed accuracy and convergence of learned solutions, identifying sensitivity to architecture and optimization dynamics.	

PUBLICATIONS

Mathews Jr., P (Electronic Communications in Probability, 2023) <i>Improved Critical Drift Estimates for the Frog Model on Trees</i>
(This project was initiated during the 2023 Polymath Jr. Program. The authors are, in alphabetical order: Emma Bailey, Yanni Bills, Feng Cheng, Osanda Chinthila, Aryan Dugar, Ali Eser, Eric Han , Chirag Kar, Matthew Junge, Viet Le, Jiaqi Liu, Zoe McDonald, Si Tang, Michael Thomas, Scott Wynn, and Eric Yu)

PRESENTATIONS

Joint Mathematics Meetings	<i>January 2024</i>
<i>San Francisco, CA</i>	
<i>Bills, Y; Cheng, F; Han, E; Le, V; Wynn, S; Yu, E</i>	
Monotonicity for the Frog Model with Drift on Trees	<i>Poster</i>
Joint Mathematics Meetings	<i>January 2024</i>
<i>San Francisco, CA</i>	
<i>Bills, Y; Cheng, F; Han, E; Yu, E</i>	
Improved Critical Drift Estimates for the Frog Model on Trees	<i>Lecture</i>

HONORS AND GRANTS

Stony Brook University	<i>2025-2026</i>
Full funding and tuition coverage for the pursuit of a Ph.D.	
Baruch Scholars	<i>2021-2023</i>
The program provides academic opportunities, supportive advisement, and four-year full-tuition scholarships for in-state students.	
Zicklin Undergraduate Honors Program	<i>2021-2023</i>
The Zicklin Undergraduate Honors Program is the only honors program at Baruch College that is focused exclusively on business education. It provides access to the minor in Advanced Business Analytics.	

ADDITIONAL EXPERIENCE

BU Directed Reading Program	<i>February 2024 - August 2024</i>
<i>Mentor: Ben Krewson, BU PhD Student</i>	
<i>Book: Nonlinear Dynamics and Chaos; Steven Strogatz</i>	
An expository project on nonlinear dynamics and chaos. A presentation was given at the end of the program. Topics covered included linear systems, existence and uniqueness of 2-dimensional flows, phase planes, limit cycles, and 2-dimensional bifurcations.	

Applied Math Project; Celestial Mechanics

March 2024 - May 2024

Professor Gabriel Ocker

Methods of Applied Math 2 Final Project

An expository work on Lagrangian mechanics from the point of view of the calculus of variations as applied to celestial mechanics, specifically the 2-body problem. Extensive research into the 2-body problem and the basis of the calculus of variations, as well as their consequences on one another, including dimensional reduction of the problem, applications, and physical consequences. Rigorous derivation of the Euler-Lagrange equations on the basis of the calculus of variations.

COURSEWORK

Graduate

Complex Analysis, ODEs, Numerical Analysis, Scientific Computing, Dynamical Systems

Atmospheric Boundary Layer, Dynamical Oceanography

Undergraduate

Methods of Applied Math 1, Methods of Applied Math 2, Modern Algebra 1, Probability, Real Analysis

RECENT WORK EXPERIENCE

Stony Brook University

August 2025 - Present

Teaching Assistant

- Mentored 200+ students through targeted weekly office hours, translating abstract theory into structured problem-solving approaches, improving student exam performance.

Prestige Institute

June 2025 - August 2025

Teacher

- Taught SAT math to high school students in a classroom setting, reinforcing fundamentals and cultivating rapid problem-solving skills; students improved practice scores by an average of 300 points.
- Delivered structured lessons and guided practice in geometry, probability, statistics, and advanced algebra, blending conceptual instruction with exam-specific strategies.

New York City Department of Youth and Community Development

June 2022 - August 2023

College Aide

- Shadowed 1 program manager and assisted them in tracking End of Year reports for community-based organizations from around New York City.
- Acted as the chief data analyst within the discretionary department for budgets and resources allocated to community-based organizations in 6 police precincts with high crime, ensuring accurate data delivery and a comprehensive overview of information.

TECHNICAL SKILLS

Computer Languages

Python, Julia, C, MATLAB, Mathematica, SageMath

Core Skills

Algorithm Development, Numerical Methods for PDEs, PINNs, Physics-Based Modeling

Tools

LaTeX, PyTorch, LAPACK, CAD (Solidworks)